

Lesson 12 Notes Logic Building Program in Java :

```
int n,r,sum=0; ✓  
System.out.println("Enter any digit:"); ✓  
n=so.nextInt(); ✓  
while(n!=0)  
{  
    r=n%10;  
    sum=sum+r;  
    n=n/10;  
}  
System.out.println("sum of the digit is:"+sum);
```

n=423 → 9
r=423%10= 3
sum=0+3=3
n=423/10=42
n=42
r=42%10=2
sum=3+2=5
n=42/10=4
n=4
r=4%10=4
sum=5+4=9
n=4/10=0
n=0

```
int n,r,sum=0; ✓  
System.out.println("Enter any digit:"); ✓  
n=so.nextInt(); ✓  
while(n!=0)  
{  
    r=n%10;  
    sum=sum+r;  
    n=n/10;  
}  
System.out.println("sum of the digit is:"+sum);
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n=42
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sum=3+2=5
n=42/10=4
n=4
r=4%10=4
sum=5+4=9
n=4/10=0
n=0

```

int n,r,rev=0;
System.out.println("Enter any digit:");
✓ n=so.nextInt();

while(n!=0)✓
{
    r=n%10;
    rev=(rev*10)+r;
    n=n/10;
}
System.out.println("Reverse is:"+rev);

```

n=423 → 324

$r=423\%10=3$
 $rev=(0*10)+3=3$
 $n=423/10=42$

n=42

$r=42\%10=2$
 $rev=(3*10)+2=32$
 $n=42/10=4$

n=4

$r=4\%10=4$
 $rev=(32*10)+4=324$
 $n=4/10=0$

n=0

```

int n,r,rev=0;
System.out.println("Enter any digit:");
✓ n=so.nextInt();

while(n!=0)✓
{
    r=n%10;
    rev=(rev*10)+r;
    n=n/10;
}
System.out.println("Reverse is:"+rev);

```

n=423 → 324

$r=423\%10=3$
 $rev=(0*10)+3=3$
 $n=423/10=42$

n=42

$r=42\%10=2$
 $rev=(3*10)+2=32$
 $n=42/10=4$

n=4

$r=4\%10=4$
 $rev=(32*10)+4=324$
 $n=4/10=0$

n=0

Palindrome Number

121

121

~~n==rev~~

~~0==121~~

```

int n,r,rev=0;
System.out.println("Enter any digit:");
n=so.nextInt();
while(n!=0)
{
    r=n%10;
    rev=(rev*10)+r;
    n=n/10;
}
System.out.println("Reverse is:"+rev);

```

~~n==rev
0==121~~

n=423 → 324

$r=423\%10=3$
 $rev=(0*10)+3=3$
 $n=423/10=42$

n=42

$r=42\%10=2$
 $rev=(3*10)+2=32$
 $n=42/10=4$

n=4

$r=4\%10=4$
 $rev=(32*10)+4=324$
 $n=4/10=0$

n=0

Palindrome Number

121

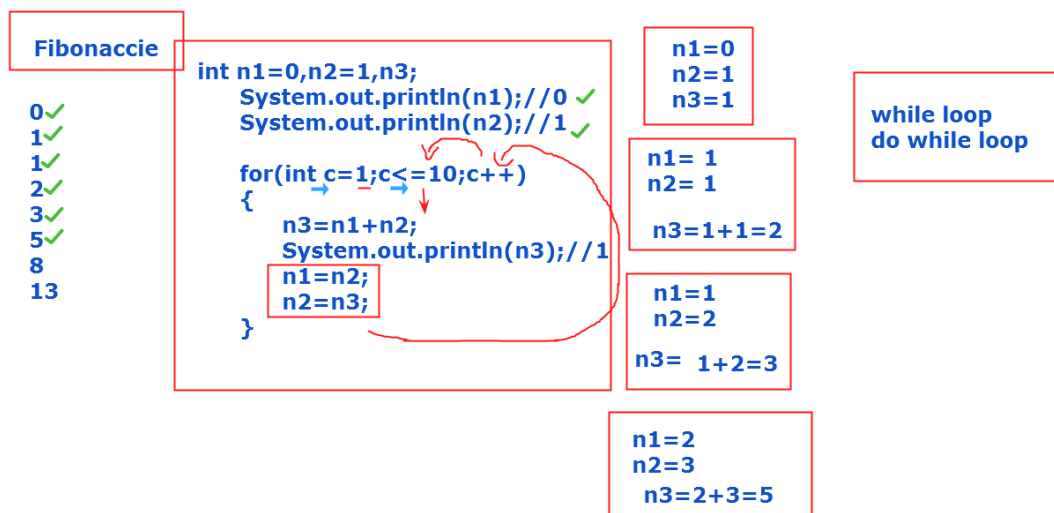
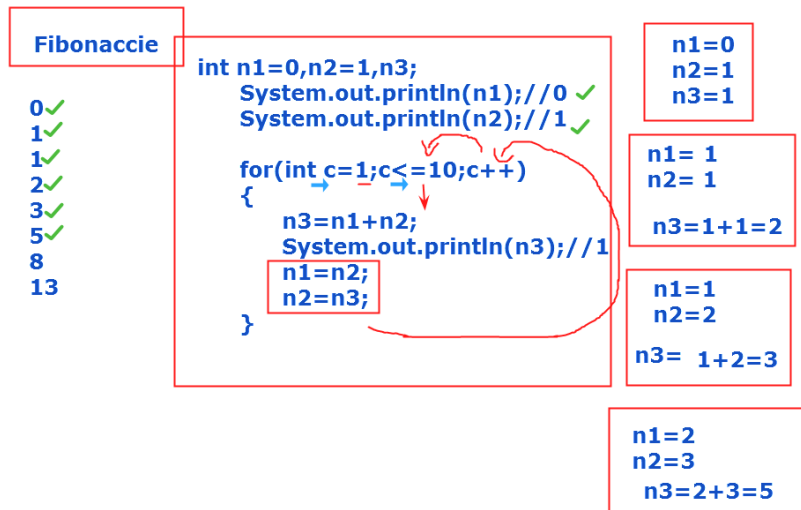
121

Armstrong number

153

$1*1*1=1$
 $5*5*5=125$
 $3*3*3=27$

153



```
import java.util.Scanner;
```

```
class A
```

```
{
```

```
    public static void main(String args[])
```

```
{
```

```
Scanner so=new Scanner(System.in);

int n,r,sum=0;

System.out.println("Enter any digit:");

n=so.nextInt();


while(n!=0)

{

    r=n%10;

sum=sum+r;

n=n/10;

}

System.out.println("sum of the digit is:"+sum);


}

}
```

```
import java.util.Scanner;

class A

{

    public static void main(String args[])
```

```
{  
    Scanner so=new Scanner(System.in);  
    int n,r,rev=0;  
    System.out.println("Enter any digit:");  
    n=so.nextInt();  
  
    while(n!=0)  
    {  
        r=n%10;  
        rev=(rev*10)+r;  
        n=n/10;  
    }  
    System.out.println("Reverse is:"+rev);  
  
}  
}
```

```
import java.util.Scanner;
```

```
class A
```

```
{
```

```
public static void main(String args[])
{
    Scanner so=new Scanner(System.in);
    int n,r,rev=0;
    System.out.println("Enter any digit:");
    n=so.nextInt();
    int x=n;
    while(n!=0)
    {
        r=n%10;
        rev=(rev*10)+r;
        n=n/10;
    }
    System.out.println("Reverse is:"+rev);

    if(x==rev)
    {
        System.out.println("Given number is a palindrome
number:");
    }
}
```

```
    else
    {
        System.out.println("Given number is not palindrome
number:");
    }

}

}
```

```
class A
{
    public static void main(String args[])
    {
        int n1=0,n2=1,n3;

        System.out.println(n1);//0
        System.out.println(n2);//1

        for(int c=1;c<=10;c++)
        {
```



```
n3=n1+n2;
```

```
System.out.println(n3);//1
```

```
n1=n2;
```

```
n2=n3;
```

```
}
```

```
}
```

```
}
```

